

THERMAL ENGINEERING – I

Time Allowed: 3 Hours

Full Marks: 70

Answer to Question No.1 is compulsory and to be answered first.

This answer is to be made in separate loose script(s) provided for the purpose.

Maximum time allowed is 45 minutes, after which the loose answer scripts will be collected and fresh answer scripts for answering the remaining part of the question will be provided.

**On early submission of answer scripts of Question No.1,
a student will get the remaining script earlier.**

Answer five questions from Group-A & B, as directed.

1. A. Answer the following questions (any ten): 10x1
- i) Which of the following is a non-renewable resource? – (a) Coal (b) Forests (c) Water (d) Wildlife.
 - ii) Which of the following is a disadvantage of most of the renewable energy sources? – (a) Highly polluting (b) High waste disposal cost (c) Unreliable supply (d) High running cost.
 - iii) Both power and manure is provided by – (a) Nuclear plants (b) Thermal plants (c) Biogas plants (d) Hydroelectric plant.
 - iv) The amount of heat required to raise the temperature of the unit mass of gas through one degree at constant volume, is called – (a) specific heat at constant volume (b) specific heat at constant pressure (c) kilo Joule (d) none of these.
 - v) An adiabatic process is one in which – (a) no heat enters or leaves the gas (b) the temperature of the gas changes (c) the change in internal energy is equal to the mechanical work done (d) all of the above.
 - vi) Work done in a free expansion process is – (a) zero (b) minimum (c) maximum (d) positive.
 - vii) The ratio of specific heat at constant pressure (C_p) and specific heat at constant volume (C_v) is – (a) equal to one (b) less than one (c) greater than one (d) none of these.
 - viii) The value of gas constant (R) in S.I. units is – (a) 0.287 J/kgK (b) 2.87 J/kgK (c) 28.7 J/kgK (d) 287 J/kg.
 - ix) With usual notation $\Delta x/kA$ is called – (a) Thermal resistance (b) Thermal conductance (c) Heat flux density (d) Thermal loading.
 - x) Pressure for Water at critical point – (a) 221.2 bar (b) 221.2 N/sq.m (c) 225.65 kgf/cm² (d) None of the above.
 - xi) The sum of internal energy (U) and the product of pressure and volume ($p.v$) is known as – (a) workdone (b) entropy (c) enthalpy (d) none of these.
 - xii) The boiling point of steam depends on – (a) Temperature (b) pressure (c) Enthalpy (d) all of the above.

B. State whether the following statements are True or False (any ten):

10x1

- i) In the first law of thermodynamics, the total energy of the system remains constant.
- ii) The kinetic energy of molecules of a gas becomes zero at absolute zero temperature.
- iii) Entropy is a point function.
- iv) The solid fuels have higher calorific value than liquid fuels.
- v) At the time of condensation heat released by steam is sensible heat.
- vi) Steam is a non homogeneous compound.
- vii) 2nd law of thermodynamics indicates the direction of heat transfer.
- viii) A black body reflects all the radiation energy.
- ix) PMM of the second kind is always possible to construct.
- x) Work done is a path function.
- xi) At triple point only steam exists.

Group-A

Answer any three questions.

- 2. Classify energy sources. Discuss disadvantages of Renewable energy. Write short notes on Geothermal Energy, Fossil Fuel, Green Energy. 2+2+6
- 3. Discuss Zeroth law of Thermodynamics. Derive Steady flow energy equation. Write down short notes on Intensive and Extensive properties, quasi static process. 2+4+4
- 4. A non flow Quasi static process (Reversible) occurs for which $p = (-3V+16)\text{bar}$, where V is volume in m^3 . What is the work done if V changes from 2 to 6 m^3 . Derive $PV^n = \text{Const.}$ (with usual notation) 4+6
- 5. Define heat engine, Refrigerator, Heat pump. Prove Reversible engines working between two fixed temperature always be more efficient than any other engine working between same two fixed temperatures. 6+4
- 6. 0.45 kg of Air at 179°C expands adiabatically and reversibly to three times its original volume and during the process there is a fall in temperature to 15.5°C . The work done during the process is 53kJ. Calculate C_p and C_v . Derive the expression of Enthalpy for constant pressure non flow process. 7+3

Group-B

Answer any two questions.

- 7. Define Dryness fraction, Triple point, Critical point, Super heated steam. Discuss phase transformation with diagram. (4x2)+2
- 8. Find the enthalpy, Internal Energy, and Entropy of 1kg of steam at a pressure of 10 bar when steam is (a) Dry saturated, (b) When steam is .75 Dry, (c) When steam is at 250°C . Use steam table. 10
- 9. Discuss Fourier's Law of Conduction. Derive an expression of conduction through a flat composite wall of several homogeneous layers. 3+7
- 10. Discuss cross flow heat exchanger. Define Grey body, Black body, Absorptivity, Stefan Boltzman law. 2+(4x2)

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On early submission of answer scripts of Question No.1, a student will get the remaining script earlier.

Answer any five questions from Group-A & B, as directed.

1. Answer the following questions (any twenty):

20x1

Choose the correct alternatives from the following:

- i) A closed system receives 60 kJ of heat but its internal energy decreases by 30 kJ. Then the work done by the system is – (a) -90 kJ, (b) -30 kJ, (c) 30 kJ, (d) 90 kJ.
- ii) The internal energy (u) of a dry steam is given by, [p is in bar, v_g is in m^3/kg , u is in kJ/kg] – (a) $u = h_g + 100 p v_g$ (b) $u = h_g - 100 p v_g$, (c) $u = h_g / 100 p v_g$ (d) $u = 100 p v_g$.
- iii) Which of the following is an intensive property of a thermodynamic system? – (a) volume, (b) temperature, (c) energy, (d) mass.
- iv) A composite wall of a furnace has 3 layers of equal thickness having thermal conductivities in the ratio of 1:2:4. What will be the temperature drop ratio across the three respective layer? – (a) 1:2:4, (b) 4:2:1, (c) 1:1:1, (d) $\log 4: \log 2: \log 1$.
- v) Entropy may be expressed as a function of – (a) pressure and temperature, (b) temperature and volume, (c) heat and work, (d) both (a) & (b).
- vi) The value of universal gas constant (R_u) is – (a) 83.14 J/kgK, (b) 8.314 J/kgK, (c) 831.4 J/kgK, (d) 8314 J/kgK.
- vii) In 100 kg wet saturated steam, mass of the water particle is 10 kg. The dryness fraction of the steam will be – (a) 0.1, (b) 1, (c) 0.85, (d) 0.9.
- viii) The change in internal energy is zero during – (a) adiabatic process, (b) polytropic process, (c) isothermal process, (d) throttling.
- ix) When gas is heated at constant pressure – (a) raises the temperature of the gas, (b) increases the internal energy of the gas, (c) does some external work during expansion, (d) both (b) & (c), (e) both (a) & (b).
- x) The general law of expansion or compression is $PV^n = \text{constant}$. The process is said to be hyperbolic if n is equal to – (a) 0, (b) γ , (c) 1, (d) ∞ .
- xi) If a heat engine gives an output of 3 kW when the input is 10000 J/sec, then the thermal efficiency of the engine will be _____.
- xii) The specific volume of water _____ when heated from 0°C . (decreases/ /increases/ first decreases, then increases / first increases, then decreases)

- xiii) The emissivity is highest for – (a) Rubber, (b) Paper, (c) iron oxide, (d) Carbon.
- xiv) The highest value of thermal conductivity is expected for – (a) Water, (b) Solid ice, (c) Melting ice, (d) Steam.

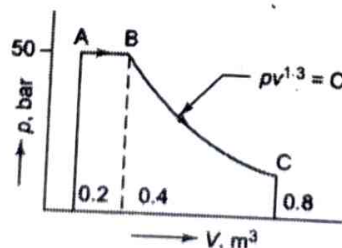
State whether the following statements are True or False:

- xv) The perpetual motion machine of first kind is possible.
- xvi) Zeroth law of thermodynamics deals with temperature.
- xvii) Ratio of two specific heats of a perfect gas is always varies with temperature.
- xviii) In isothermal process change in internal energy is zero.
- xix) Efficiency of any heat engine cycle is always 100%.
- xx) Entropy of universe is increasing.
- xxi) Quasistatic process is a reversible process.
- xxii) Heat is low grade energy.

Group-A

Answer any three questions.

2. a) Write down the advantages and disadvantages of Geothermal Energy.
b) Write short notes on: - Solar water heater, Biogas plant. 4+6
3. a) Differentiate between extensive and intensive properties with example.
b) Prove that heat transfer is a path function.
c) Show that the work done for an isothermal process from the state 1 to state 2 is given by $W_{1-2} = p_1 v_1 (\ln p_1 - \ln p_2)$, where each term has its usual meaning. 3+3+4
4. a) What is steady flow process? Write the steady flow energy equation for a single stream entering and a single stream leaving a control volume and explain the various terms of it.
b) A gaseous system undergoes three quasi-static processes in sequence. The gas initially at 5 bar and 0.01 m^3 is expanded at constant pressure. It is then further expanded according to the relationship $p v^2 = \text{constant}$ to 2 bar and 0.025 m^3 . The gas is then returned to its initial state during which $p v = \text{constant}$. Sketch the cycle in the P-V diagram. Calculate the work done in each case. (1+4)+(1+4)
5. a) Determine the total work done by a gas system following an expansion process as shown in Figure.



- b) A mass of 8 kg gas expands within a flexible container so that the p-v relationship is of the form $p v^{1.2} = \text{constant}$. The initial pressure is 1000 kPa and the initial volume is 1 m^3 . The final pressure is 5 kPa. If the specific internal energy of the gas decreases by 40 KJ/Kg, find the heat transfer in magnitude and direction. 5+5
6. a) Prove that for an ideal gas, change in entropy, $S_2 - S_1 = c_v \ln \left(\frac{T_2}{T_1} \right) + R \ln \left(\frac{v_2}{v_1} \right)$.
b) An inventor claims to have heat engine which is capable of developing 9 kW while working between the temperature limits of 30°C and -40°C . It receives only 1000 kJ/min of heat. Discuss the possibility of the claim. 5+5

7. a) Establish the relation: $(C.O.P.)_{HP} = (C.O.P.)_R + 1$
 b) A mass of 50 kg/min enters the control volume of a steady flow system at 2 bar and 100°C and an elevation of 100 m above the datum. The same mass leaves the control volume at 150 m elevation with a pressure of 10 bar and temperature of 300°C. The entrance and exit velocities are 2400 m/min and 1200 m/min respectively. During the process, 50,000 kJ/hr of heat are transferred to the control volume and the rise in the enthalpy is 8 kJ/kg. Calculate the power developed. 3+7
8. a) State Kelvin-Planck and Celsius statements related to 2nd law of thermodynamics.
 b) 10 kg of air is compressed adiabatically from 1 bar pressure and 18°C to 7 bar and then cooled at constant pressure to original temperature. The air is then expanded adiabatically to the original pressure. [Take $\gamma = 1.4$ and $C_p = 1.005 \text{ kJ/kgK}$]. Find – (i) The temperature after compression, (ii) The final temperature and (iii) Heat extracted during constant pressure cooling. 4+6

Group-B

Answer any two questions.

9. a) Describe the working principle of Throttling calorimeter with neat sketch.
 b) A rigid closed tank of volume 3 m³ contains 5 kg of wet steam at a pressure of 200 kPa. The tank is heated until the steam becomes dry saturated. Determine the final pressure and the heat transfer to the tank. 4+6
10. a) Define dryness fraction of steam. What do you mean by degree of superheat?
 b) Write down the advantages of using Superheated steam.
 c) In a throttling calorimeter, the steam is admitted at a pressure of 10 bar. If it is discharged at atmospheric pressure and 110°C after throttling, determine the dryness fraction of steam. Assume specific heat as steam as 2.2 kJ/kg K. 4+3+3
11. a) State Fourier's law of heat conduction.
 b) A composite wall consisting of three different layers in perfect thermal contact is shown in sketch below. The outer surfaces to the extreme left and right are kept at temperatures 400°C and 50°C respectively. The thickness L_i for $i = 1, 2, 3$ of each layer are given as 5cm, 10cm and 15 cm and the thermal conductivity k_i for $i = 1, 2, 3$ of each layer are given as 20W/m°C, 50W/m°C and 100W/m°C respectively. Determine the heat transfer rate per square meter across this composite layer by assuming one dimensional heat flow and using the thermal resistance concept. 3+7
12. a) Distinguish between – (i) Black body and gray body, (ii) Absorptivity and emissivity of a surface.
 b) Deduce the formula of overall heat transfer co-efficient for a flat composite wall, taking into consideration the convective heat transfer. 4+6

March 2021

THERMAL ENGINEERING – I

Time Allowed: 3 Hours

Full Marks: 70

Answer to Question No.1 is compulsory and Answer any five questions from the rest.

1. Answer the following questions (any twenty):

1×20

State True or False:

- i) During throttling process enthalpy remains constant.
- ii) H-S diagram for water steam is called Mollier diagram.
- iii) Superheated steam behaves like a perfect gas.
- iv) During heat transfer by radiation no medium is required.
- v) Intensive properties per unit mass are extensive properties.
- vi) A process with identical end states is called a cycle.
- vii) PMM-II violates the first law of thermodynamics.

Choose correct answer from the given alternatives:

- viii) Which of the following relation is correct? -
a) $C_p/C_v=R$ b) $C_p+C_v=R$ c) $C_p/C_v=\gamma$ d) $C_p-C_v=\gamma$
- ix) For same temperature limit which relation will be correct? -
a) $1 - \text{COP} = \eta_{\text{HE}}$ b) $1 + \text{COP} = \eta_{\text{HE}}$
c) $\text{COP} - \eta_{\text{HE}} = 1$ d) None of these
- x) The work transfer for isochoric process will be -
a) $P_1 V_1 \ln P_1/P_2$ b) $(P_1 V_1 - P_2 V_2)/(N-1)$
c) Zero d) None of these
- xi) For which of the following processes Heat Transfer will be zero? -
a) Adiabatic b) Isochoric c) Isobaric d) None of these
- xii) Dryness fraction of wet saturated steam will be -
a) >1 b) <1 c) $=1$ d) 0
- xiii) Which of the following relation is true? -
a) $1 = \alpha + \beta + \gamma$ b) $\alpha + \beta - 2\gamma = 2$ c) $\alpha - \beta - \gamma = 0$ d) $\alpha\beta = \gamma$
- xiv) Which of the following is renewable source of energy? -
a) Water b) Coal c) Natural Gas d) None of these
- xv) Turbine are used in following power plant -
a) Solar b) Thermal c) Nuclear d) Both b & c
- xvi) A cup of tea place on a table is an example of -
a) Open system b) Closed System
c) Isolated system d) None of these
- xvii) Which will be is unit of Power given below? -
a) KJ/Kg b) KJ/hr c) Cal/Kg d) KJ-hr

Fill in the blanks:

- xviii) The value of universal gas constant is _____.
xix) The work done in free expansion process is _____.
xx) The locus of saturated liquid line and saturated vapor line meets at _____.
xxi) The latent heat of steam at atmospheric pressure is _____.
xxii) The thermal conductivity of solid metals _____ with rise in temperature.
xxiii) Summation of internal energy and _____ is called enthalpy.
xxiv) Atmospheric pressure is equal to _____ bar.
xxv) Kelvin-Planks law deals with _____.
2. Define heat engine, Refrigerator, Heat pump. Prove Reversible engines working between two fixed temperature always be more efficient than any other engine working between same two fixed temperatures. 6+4
3. 0.45 kg of Air at 179°C expands adiabatically and reversibly to three times its original volume and during the process there is a fall in temperature to 15.5 °C. The work done during the process is 53 kJ. Calculate C_p and C_v . Derive the expression of Enthalpy for constant pressure non flow process. 7+3
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THERMAL ENGINEERING-I

Time Allowed: 3 Hours

Full Marks: 60

Answer the following questions from Group-A, B & C as directed.

Group A

1. Choose the correct answer from the alternatives (any ten):

1 x 10

- i. Heat pump works on the principle of - a) Kelvin plank Statement, ~~b) Clausius statement~~, c) Zeroth law of thermodynamics, d) energy equation of flow process.
- ii. In steady flow process - ~~a) the mass flow rate is constant~~, b) the heat transfer rate is constant, c) the work transfer rate is constant, d) all of these.
- iii. Superheated vapour behaves - a) Exactly as gas, b) As steam, c) As ordinary vapour, d) Approximately as a gas.
- iv. In an economiser - a) Feed water is heated by the heat extracted from flue gas, b) Primary air is heated by the heat extracted from the flue gas, c) Suspended solids are separated from the flue gas using electrostatic forces, d) Feed water is pumped to the de-aerator.
- v. Dryness fraction of a steam containing 20% of water vapour by mass is - a) 0.2, b) 80, c) 0.8, d) 20.
- vi. Heat and work are - (a) intensive properties, (b) extensive properties, (c) point functions, ~~(d) path functions~~.
- vii. Throttling of steam during dryness fraction determination by throttling calorimeter is essentially - a) isenthalpic process, b) isentropic process, c) isochoric process, d) isobaric process.
- viii. A series of operations, which take place in a certain order and restore the initial condition is known as- ~~a) reversible cycle~~, b) irreversible cycle, c) thermodynamic cycle, d) None of these.
- ix. The value of C_p/C_v for air is - ~~a) 1~~, b) 1.4, c) 1.8, d) 2.3.
- x. Mollier chart is also known as - a) T-S diagram b) p-T diagram c) p-V diagram d) h-S diagram
- xi. Which of the following statements is wrong? - a) Mechanical draught reduces the height of chimney., b) Natural draught reduces the fuel consumption., c) A balanced draught is a combination of induced and forced draught., d) All of the statements.
- xii. For a reversible adiabatic process, the change in entropy is - a) minimum, b) maximum, c) unity, ~~d) zero~~.
- xiii. According to Clausius statement of the second law of thermodynamics, the coefficient of performance of a refrigerator is never- (a) infinite, (b) greater than unity, (c) unity, ~~(d) less than unity~~.
- xiv. A device used in boiler to control the flow of steam from the boiler to the main pipe and shut off the steam completely when required, is known as - a) blow off cock, b) fusible plug, ~~c) stop valve~~, d) economiser.
- xv. Water tubes in a Babcock and Wilcox boiler are - ~~a) Horizontal~~, b) Vertical, c) Inclined, d) inclined.

2. Fill in the blanks (any ten):

1x10

- i. PMM2 violates _____ law.
- ii. Work is a _____ function.
- iii. Quantity of steam generated by a boiler at full load is called _____ of the boiler.
- iv. COP of heat pump is _____ than COP of refrigerator.
- v. First law of thermodynamics refers to the conservation of _____.
- vi. Thermal efficiency of a Carnot engine working between temperature levels T_1 and T_2 ($T_1 > T_2$) is given by _____.
- vii. Change of internal energy is zero during _____ process.
- viii. The unit of pressure in S.I. unit is _____.
- ix. One example of vertical fire tube boiler is _____.
- x. In La-mont boiler there is _____ circulation of water and steam.
- xi. For reversible adiabatic process _____ remains constant.
- xii. At critical point, the enthalpy of vaporisation i.e h_{fg} , is equal to _____.
- xiii. Total heat before throttling is _____ to total heat after throttling.
- xiv. As per Charles law, volume of given mass of a perfect gas varies _____ with its absolute temperature, when the absolute pressure remains constant.
- xv. Absolute zero temperature is _____.

3. Answer the following questions (any ten):

1x10

- i. State Boyle's law.
- ii. What is meant by quasi-static process in thermodynamics?
- iii. Differentiate universal gas constant and characteristic gas constant.
- iv. Give two examples of boiler accessories.
- v. Define Evaporation Ratio of a boiler.
- vi. Give one example of an open system.
- vii. Define Dryness fraction.
- viii. State Avogadro's law.
- ix. What is an isolated system?
- x. Write the value of characteristic gas constant?
- xi. What is superheated steam?
- xii. Under what circumstances, steam can be treated as an ideal gas?
- xiii. Why work is treated as high grade energy?
- xiv. Define a pure substance?
- xv. State the use of a calorimeter.

Group B

4. Answer the following questions (any six):

2x6

- i. Calculate the work done by a gas in an isothermal process from A to B. $P_A = 1 \text{ Pa}$, $V_A = 3 \text{ m}^3$ and $P_B = 3 \text{ Pa}$.
- ii. How does fusible plug work in fire tube boiler?
- iii. Explain in brief the concept of flow work.
- iv. What do you mean by entropy of wet steam?
- v. State Kelvin Planck statement.
- vi. Briefly state the function of an air-preheater.
- vii. Show different thermodynamic processes on P-V and T-S diagram.
- viii. Define a perfect gas.
- ix. Why circulating pump is required for high pressure boilers?
- x. What is latent heat of vaporisation?

Group C

5. Answer the following question (any one):

1x6

- i. A Carnot heat engine receives heat from a reservoir at 900°C at a rate of 800 kJ/min and reject the waste heat to the ambient air at 27°C . The entire work output of the engine is used to drive a refrigerator that removes heat from the refrigerated space at -5°C and transfers it to the same ambient air at 27°C . What will be the maximum rate of heat removal (in kJ/min) from the refrigerated space?
6
- ii. Define the following with examples: a) Open system, b) Closed system, c) Isolated system.
2+2+2
- iii. State First Law of Thermodynamics. What are the limitations of this law? Give an example of application of the law.
2+3+1

6. Answer the following question (any one):

1x6

- i. Show that $C_p - C_v = R$.
6
- ii. Prove that during adiabatic process, work done is given by $W = (P_1 V_1 / \gamma - 1) \{1 - P_2 V_2 / P_1 V_1\}$.
6
- iii. Steam expands isentropically in a nozzle from 1 MPa , 250°C to 10 kPa . The steam flow rate is 1 kg/s . Neglecting the K.E of steam at inlet to the nozzle, find the velocity of steam at exit from the nozzle and the exit area of the nozzle?
6

7. Answer the following question (any one):

1x6

- i. In a separating and throttling calorimeter the total quantity of steam passed was 40 kg and 2.2 kg of water was collected from separator. Steam pressure before throttling was 1.5 MPa and temperature and pressure after throttling are 120°C and 105 kPa . Determine the dryness fraction of steam before entering to calorimeter. Specific heat of superheated steam may be considered as 2.09 kJ/kg.K .
6
- ii. Draw a simple neat sketch of Cochran boiler, Label all its components and briefly mention some common features of the boiler.
6
- iii. What are boiler mountings and accessories? What is the difference between them? Give 3 examples each of boiler mountings and accessories.
2+2+2

THERMAL ENGINEERING-I*Time Allowed: 2.5 Hours**Full Marks: 60*

Answer to Question No. 1 of Group A is compulsory and to be answered first. This answer is to be made in separate loose script(s) provided for the purpose. Maximum time allowed is 30 minutes, after which the loose answer scripts will be collected and fresh answer scripts for answering the remaining part of the question will be provided. On early submission of answer scripts of Question No. 1, a student will get the remaining script earlier.

Answer any Five (05) Questions from Group B.

Group A

1. Choose the correct answer from the given alternatives (any twenty): 1x20
 - a. The value of $n = 1$ in the polytropic process indicates it to be – (a) adiabatic process (b) isobaric process (c) isothermal process (d) isentropic process
 - b. Measurement of temperature is based on _____ law of thermodynamics – (a) zeroth (b) first (c) second (d) third
 - c. An open system is one in which _____. (a) Mass does not cross boundaries of the system, though energy may do so (b) Neither mass nor energy crosses the boundaries of the system (c) Both energy and mass cross the boundaries of the system (d) Mass crosses the boundary but not the energy
 - d. The first law of Thermodynamics relates to – (a) Conservation of mass (b) Conservation of momentum (c) Conservation of internal energy (d) Conservation of energy
 - e. Properties of substances like pressure, temperature and density, in thermodynamic coordinates are – (a) path functions (b) point functions (c) cyclic functions (d) real functions
 - f. “No process is possible whose sole result is the transfer of heat from a colder object to a hotter object” this statement is related to the – (a) zeroth law of thermodynamics (b) first law of thermodynamics (c) Kelvin Planck statement (d) Clausius statement
 - g. A machine which can supply mechanical work continuously without consumption of any energy is called as – (a) PMM1 (b) PMM2 (c) PMM3 (d) all the above
 - h. Boyle’s law relates to – (a) Constant volume (b) Constant pressure (c) Constant temperature (d) none of the above
 - i. For a perfect gas the specific heat at constant pressure C_p and that at constant volume C_v are related as – (a) $C_p = C_v$ (b) $C_p = C_v + R$ (c) $C_p = C_v - R$ (d) $C_p = C_v + \log R$
 - j. The statement that molecular weights of all gases occupy the same volume is known as – (a) Dalton’s law (b) Joule’s law (c) Boyle’s law (d) Avogadro’s hypothesis
 - k. The conditions at which a real gas behaves as an ideal gas – (a) high pressure (b) low temperature (c) low pressure (d) at any pressure
 - l. When a gas is compressed adiabatically, its temperature – (a) remains constant (b) increases (c) decreases (d) none of the above
 - m. The point that connects the saturated liquid line and saturated vapour line is called – (a) saturated point (b) critical point (c) triple point (d) superheated point

- n. Dryness fraction of dry saturated steam on heating – (a) increases (b) decreases (c) remain same (d) cannot predict
- o. Latent heat of vapourisation of steam – (a) increases with the increase of pressure (b) decreases with the increase of pressure (c) remains constant at all pressures (d) latent heat doesn't depend upon pressure
- p. Measure the dryness fraction of a very wet steam with the help of – (a) separating calorimeter (b) throttling calorimeter (c) combined separating and throttling calorimeter (d) bomb calorimeter
- q. The critical pressure of water is – (a) 7.38 MPa (b) 22.12 MPa (c) 100 MPa (d) 5.04 MPa
- r. The isentropic process on the Mollier chart is represented by – (a) horizontal line (b) vertical line (c) inclined line (d) curved line
- s. Equivalent evaporation is the amount of water evaporated in a boiler from and at – (a) 0°C (b) 100°C (c) saturation temperature at given pressure (d) room temperature
- t. Cochran boiler is a – (a) horizontal fire tube boiler (b) horizontal water tube boiler (c) vertical fire tube boiler (d) vertical water tube boiler
- u. A fusible plug is fitted in small boilers in order to – (a) control steam flow (b) extinguish fire if water level in boiler falls below alarming limit (c) remove ash (d) all the above <https://www.wbsctonline.com>
- v. The draught in boiler may be produced by – (a) mechanical fan (b) chimney (c) a steam jet (d) all of these
- w. _____ is NOT a boiler mounting. – (a) Blow off cock (b) Feed check valve (c) Economiser (d) Safety valve
- x. The water tubes in a Babcock and Wilcox boiler are – (a) horizontal (b) vertical (c) curved (d) inclined
- y. The maximum heat loss in a boiler occurs due to – (a) moisture in fuel (b) dry flue gases (c) steam formation (d) unburnt carbon

Group B

2.
 - a) A cylinder contains 0.45 m^3 of a gas at $1 \times 10^5\text{ N/m}^2$ and 80°C . The gas is compressed to a volume of 0.13 m^3 , the final pressure being $5 \times 10^5\text{ N/m}^2$. Determine: (i) The mass of gas; (ii) The value of index 'n' for compression; (iii) The increase in internal energy of the gas; (iv) The heat received or rejected by the gas during compression. Take $\gamma = 1.4$, $R = 294.2\text{ J/kg}^{\circ}\text{C}$.
 - b) Make a comparison of ideal gas with a real gas. 6+2
3.
 - a) Draw a Mollier diagram showing different lines.
 - b) The following observations were recorded in an experiment with a combined separating and throttling calorimeter:
 Pressure in the steam main: 15 bar
 Mass of water drained from the separator: 0.55 kg
 Mass of steam condensed after passing through the throttle valve: 4.20 kg
 Pressure and temperature after throttling: 1 bar, 120°C
 Using steam tables, evaluate the dryness fraction of the steam in the main, and state with reasons, whether the throttling calorimeter alone could have been used for this test. 2+6
4.
 - a) State Clausius statement of second law of thermodynamics. With the help of block diagrams, show the COP of refrigerator and COP heat pump.
 - b) A turbine operates under steady flow conditions, receiving steam at the following state: Pressure 1.2 MPa, temperature 188°C , enthalpy 2785 kJ/kg , velocity 33.3 m/s and elevation 3 m. The steam leaves the turbine at the following state: Pressure 20 kPa, enthalpy 2512 kJ/kg , velocity 100 m/s , and elevation 0 m. Heat is lost to the surroundings at the rate of 0.29 kJ/s . If the rate of steam flow through the turbine is 0.42 kg/s , calculate the power output of the turbine in kW. 3+5

5. a) Explain the steam generation process with the help of p-v and T-s diagram.
b) Define wet steam, dry saturated steam and superheated steam. 5+3
 6. a) What do you mean by system? Classify systems and explain each system.
b) What do you mean by properties of a system? Define intensive properties and extensive properties with proper examples. 4+4
 7. a) Prove that adiabatic process follows the law $PV^\gamma = \text{const.}$
b) A cylinder contains 0.02 m^3 of air at 1.66 MPa and 50°C . The air expands isothermally to 104 KPa . Determine: i) Work transfer, ii) Heat transfer, iii) Final volume. 4+4
 8. a) List the different sources of heat losses in a boiler.
b) The following readings were taken during the test on a boiler for a period of 1 hour: steam generated 5000 kg ; coal burnt 700 kg ; CV of coal 31402 kJ/kg and quality of steam 0.92 . If the boiler pressure is 1.2 MPa and feed water temperature is 45°C , calculate the boiler equivalent evaporation and thermal efficiency. [using Steam Table] 2+6
- 1.2 × 10
9. a) Explain the construction and working principle of Lamont boiler with a neat sketch.
b) Distinguish between fire tube boiler and water tube boiler. 5+3
 10. a) Derive the expression, $h = 353H \left[\frac{1}{T_a} - \left(\frac{m+1}{m} \right) \frac{1}{T_f} \right]$ of natural draught in mm of water column.
(All the symbols have their usual meanings)
b) List at least three mountings and three accessories used in boiler. 5+3

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DECEMBER 2024
THERMAL ENGINEERING-I

3247848

Time Allowed: 3 Hours

Full Marks: 60

Answer to Question No. 1 of Group A must be written in the main answer script. In Question No. 1, out of 2 marks for each MCQ, 1 mark is allotted for right answer and

1 mark is allotted for correct explanation of the answer.

Answer any Five (05) Questions from Group-B.

GROUP-A

1. Choose the correct answer from the given alternatives and explain your answer (any ten): $2 \times 10 = 20$

- i. Cochran boiler is a
(a) horizontal fire-tube boiler, (b) horizontal water-tube boiler,
(c) vertical water-tube boiler, (d) vertical fire tube boiler.
- ii. With increase of pressure, latent heat of evaporation
(a) increases, (b) decreases, (c) remain constant (d) may increases or decreases
- iii. Superheated vapour behaves
(a) Exactly as gas, (b) as steam, (c) as ordinary vapour, (d) approximately as gas.
- iv. Measurement of temperature is based on _____ law of thermodynamics.
(a) zeroth, (b) first, (c) second, (d) third.
- *****
- v. Boyle's law relates to
(a) constant volume, (b) constant pressure, (c) constant temperature, (d) none of these.
- vi. Water tube boilers are those in which
(a) flue gases pass through tubes and water around it, (b) water passes through the tubes and flue gases around it, (c) work is done during adiabatic expansion, (d) change in enthalpy.
- vii. The enthalpy of a substance is defined as
(a) $H = U - PV$, (b) $H = U + PV$, (c) $H = -U + PV$, (d) $H = -U - PV$.
- viii. When water is heated with the rise of temperature, it consumes
(a) latent heat, (b) sensible heat, (c) both (a) and (b) (d) none of these.
- ix. Heat is
(a) intensive properties, (b) extensive properties, (c) both intensive and extensive properties, (d) can't say



3247848

- x. The temperature at which the Celsius and Fahrenheit scales read the same is
(a) 60° (b) -60° (c) -40° (d) 40°
- xi. The maximum heat loss in a boiler occurs due to
(a) moisture in fuel, (b) dry flue gases, (c) steam formation, (d) unburnt carbon.
- xii. Dryness fraction(x) of superheated steam is
(a) equal to 0, (b) greater than 1, (c) less than 1, (d) equal to 1.
- xiii. The function of Blow down valve of a boiler is
(a) to remove sediments (b) to build sediments (c) to remove flue gas (d) to remove ash
- xiv. The natural draught in boiler is produced by
(a) steam jet (b) centrifugal fan (c) chimney (d) both (a) and (b)
- xv. An economiser is installed in a boiler primarily to -
(a) superheat the steam (b) reduce fuel consumption
(c) increase steam pressure (d) all of these

GROUP-B

2. a) State First law of thermodynamics. What are the limitations of this law?
b) Deduce the steady flow energy equation. [3+5]
- ***
3. a) Define with brief explanation: PMM-I and PMM-II
b) Define the following with example: (i) system, surroundings and boundary, (ii) intensive and extensive properties [4+4]
4. a) State Boyle's law and Charle's laws.
b) Show that $C_p - C_v = R$. [3+5]
5. a) What is universal gas constant?
b) 0.2 m^3 of an ideal gas at a pressure of 2 MPa and 600 K is expanded isothermally to 5 times the initial volume. It is then cooled to 300 K at constant volume and then compressed back polytropically to its initial state. Determine the net work done during the cycle. [2+6]
6. a) Deduce the expression for work done during adiabatic process.
b) Write shortly on (i) Dryness fraction of steam (ii) Air preheater of boiler [4+(2+2)]



3247848

7. a) What is meant by critical point? Draw the outline of Mollier chart and show critical point on it.
b) How much heat is needed to convert 5kg of water at 45°C into steam at pressure of 4 bar and 80 per cent dry? Take specific heat of water as 4.187 kJ/kg-K .
[(2+2)+4]
8. a) Differentiate between fire tube boiler and water tube boiler.
b) What is feed water? What are the necessities of boiler feed water treatment?
c) Define equivalent evaporation.
[3+(1+2)+2]
9. a) Explain with neat sketch the construction and working principle of Cochran boiler.
b) A boiler uses 18 kg air per kg of fuel. The mean temperature of chimney gases is 315°C and that of outside air 27°C . Determine the minimum height of chimney required to produce a draught of 20 m.
[5+3]



THERMAL ENGINEERING - I

Time Allowed: 2.5 Hours

Full Marks: 60

Answer to Question No. 1 of Group A must be written in the main answer script. In Question No. 1, out of 2 marks for each MCQ, 1 mark is allotted for right answer and 1 mark is allotted for correct explanation of the answer.

Answer any Five (05) Questions from Group-B.

GROUP-A

1. Choose the correct answer from the given alternatives and explain your answer (any ten): $2 \times 10 = 20$

- ② i) Which of the following is the extensive property of a thermodynamic system?
 a) pressure ☒ b) volume c) temperature d) density
- ① ii) In thermodynamics, heat and work are -
 a) point functions ☒ b) path functions c) both point and path functions d) non of these
- iii) Which of the following represent an open system?
 a) an air-tight container ☒ b) Pressure cooker c) a beaker full of water d) thermos flask
- iv) The work done is minimum in the following gas process.
 a) isobaric b) isochoric c) isothermal d) adiabatic
- ① v) Which of the following law of thermodynamics is used to understand the concept of energy conservation?
 a) zeroth law ☒ b) first law c) second law d) none of these
- vi) An ideal gas is one which obeys the law -
 a) $pV=RT$ b) $p = \frac{vR}{T}$ c) $pV = \frac{R}{T}$ d) none of these
- vii) A perfect gas at 300 K is heated at constant pressure till its volume is double. The final temperature is -
 a) 150 K b) 450 K c) 600 K d) No change of Temperature
- viii) At critical point, the enthalpy of vaporisation is -
 a) dependent on temperature only b) maximum c) minimum ☒ d) zero
- ix) The dryness fraction (x) of fully wet steam is -
☒ a) $x = 0$ b) $x = 1$ c) $0 < x < 1$ d) $x > 1$
- x) If the enthalpy of saturated water and dry saturated steam at certain conditions are 1185 kJ/kg and 2790 kJ/kg respectively, then the enthalpy of wet steam of dryness fraction 0.5 is -
 a) 1185 kJ/kg b) 3975 kJ/kg c) 2580 kJ/kg d) 2790 kJ/kg
- ① xi) In a fire-tube boiler, the water is in -
 a) tube ☒ b) shell c) all of these d) non of these
- ① xii) The natural draught in boiler is produced by
 a) steam jet b) centrifugal fan ☒ c) chimney d) both (a) and (b)
- ① xiii) An economiser is installed in a boiler primarily to -
 a) superheat the steam b) reduce fuel consumption c) increase steam pressure d) all of these

- ① xiv) The amount of water evaporated 'from and at 100 °C' into dry saturated steam at standard atmospheric pressure is called-
- a) evaporative capacity b) generation factor c) boiler horse power d) equivalent evaporation
- ① xv) The safety device used to protect the boiler when the water level falls below the minimum level in the boiler is-
- a) fusible plug b) safety valve c) blow-off cock d) water level indicator

GROUP-B

- ② 2. a) State first law of thermodynamics and specify its applications.
- b) A nozzle is a device for increasing the velocity of a steadily flowing stream. At inlet to a certain nozzle, the fluid parameters are: Enthalpy is 2850 kJ/kg; velocity is 50 m/s; area is 0.1 m² and specific volume is 0.18 m³/kg. At the discharge end, the Enthalpy is 2650 kJ/kg and the specific volume is 0.49 m³/kg. The nozzle is horizontal and there is negligible heat loss from it. Find the velocity of fluid at exit from the nozzle, mass flow rate and the exit area of the nozzle. [3+5]
3. a) What is thermodynamic system? Classify it and briefly describe them with examples.
- b) What is perpetual motion machine of 2nd kind (PMM2)? Write the efficiency of PMM2. Why such a machine cannot be constructed in actual practice? [(1+3) + (2+1+1)]
- ① 4. a) State the differences between real gas and ideal gas. Under which condition does a real gas behave most like an ideal gas? State the condition at which steam behaves almost like a gas.
- b) Show that $C_p - C_v = R$ [(2+1+1) + 4]
5. a) Derive an expression for the work done during an isothermal process of an ideal gas.
- b) A quantity of gas has a volume of 0.14 m³, pressure 1.5 bar and temperature 100°C. If the gas is compressed at constant pressure, until its volume becomes 0.112 m³, determine (i) the temperature at the end of compression and (ii) work done in the process. [4 + (2+2)]
- ② 6. a) List out the advantages of superheated steam.
- b) Determine the quantity of heat required to convert 1 kg of water at 30 °C to steam at a pressure of 6 bar under the following conditions:
- when the steam is dry saturated;
 - when the steam is wet having a dryness fraction 0.85; and
 - when the steam is superheated to 250°C, Take the specific heat of water as 4.2 kJ/kg K and that of the superheated steam as 2.25 kJ/kg K. [2+6]
- ② 7. a) Explain the terms 'critical point' with respect to steam generation.
- b) Write short notes on 'Mollier diagram'.
- ② c) A boiler evaporates 3.6 kg of water per kg of coal into dry saturated steam at 10 bar. The feed water temperature is 32° C. Find the equivalent evaporation from and at 100° C. [2 + 3 + 3]

⑥ 8. a) Classify boilers based on (i) relative position of hot gas and water, (ii) method of circulation of water.
Give example of each.

b) Describe with a neat sketch the working principle of Babcock & Wilcox boiler. [3 + 5]

9. a) State the functions of safety valve and blow-off cock employed in boiler.

b) State the advantages of artificial draught system over natural draught system.

c) The following data pertain to a steam power plant:

Height of chimney = 30 m ;

Draught produced 16.5 mm of water gauge ;

Temperature of flue gas = 360°C ;

Temperature of boiler house = 28°C ;

Determine the quantity of air used per kg of fuel burnt in the boiler. [2 + 2 + 4]

THERMAL ENGINEERING -I

Time Allowed: 2.5 Hours

Full Marks: 60

Answer to Question No. 1 of Group A must be written in the main answer script. In Question No. 1, out of 2 marks for each MCQ, 1 mark is allotted for right answer and 1 mark is allotted for correct explanation of the answer.

Answer any Five (05) Questions from Group-B.

GROUP-A

1. Choose the correct answer from the given alternatives and explain your answer (any ten): 2×10=20

- (i) First law of thermodynamics says about
(a) Quality of energy (b) quantity of energy (c) both of (a) & (b) (d) none of these
- (ii) Which of the following is the first law for a closed system undergoing a cycle?
(a) $\oint dW = \oint dQ$ (b) $\oint dW = -\oint dQ$ (c) $\oint dW = \oint dQ$ (d) None of the mentioned
- (iii) A heat engine produces an output of 12 kW by rejecting 23 kJ/sec of heat to a sink at a particular temperature. The efficiency of the engine will be (a) 33.4 % (b) 65.7% (c) 34.3% (d) 52.17%
- (iv) In a throttling device, what do we get as SFEE when changes in PE and KE are taken zero?
(a) $dQ/dm \neq 0$ (b) $dW/dm \neq 0$ (c) $h_1 = h_2$ (d) none of the mentioned
- (v) Coefficient of performance(COP) is defined as
(a) heat leakage/work input (b) work input/heat leakage
(c) refrigeration effect/work input (d) work input/latent heat of condensation
- (vi) What is the dryness fraction (x) for saturated water, when water just starts boiling?
(a) x = 0 (b) x = 1 (c) x = 0.9 (d) x = 0.5
- (vii) Entropy is a
(a) path function, intensive property (b) path function, extensive property
(c) point function, intensive property (d) point function, extensive property
- (viii) Triple point of a pure substance on P-V diagram is represented by a
(a) point (b) line (c) curve (d) triangle
- (ix) At a pressure below the triple point line,
(a) the substance cannot exist in the liquid phase
(b) the substance when heated transforms from solid to vapour
(c) both of the mentioned (d) none of the mentioned
- (x) The amount of heat transferred to convert unit mass of solid to vapour or vice versa is called as
(a) latent heat of vapourization (b) latent heat of fusion
(c) latent heat of sublimation (d) specific heat
- (xi) The function of a nozzle is to?
(a) decrease the velocity of a fluid at the cost of its pressure gain
(b) increase the velocity of a fluid at the cost of its pressure drop
(c) increase the velocity of a fluid and also its pressure
(d) none of the mentioned.
- (xii) A closed system receives 60 kJ of heat but its internal energy decreases by 30 kJ. Then the work done by the system is (a) - 90 kJ (b) -30 kJ (c) 30 kJ (d) 90 kJ
- (xiii) Mollier diagram is a plot of
(a) temperature and entropy (b) enthalpy and entropy
(c) pressure and enthalpy (d) pressure and volume
- (xiv) What is the function of economizer?

- (a) It is used to heat the feed water (b) It is used to improve the quality of steam
 (c) It is used to heat air being supplied for combustion
 (d) It separates suspended water particles from steam
- (xv) A mass of gas is compressed in a quasi-static process from 80 kPa, 0.1 m^3 to 0.4 MPa, 0.03 m^3 in a process following $PV^n = \text{const.}$ The index of compression is
 (a) 1.23 (b) 1.337 (c) 1.4 (d) 1.46

GROUP-B

Answer any Five (05) questions.

2. a) Define the following terms related to a thermodynamic system: Path of a process, Extensive property and & Point function. ①
 What do you understand by property of a system? $[(1 \times 3) + 1]$
- b) Prove that for a reversible adiabatic process, $PV^\gamma = \text{constant}$. ① (4)
3. a) A turbine operates under steady flow conditions, receiving steam at the following state: Pressure 1.2 MPa, temperature 188°C , enthalpy 2785 kJ/kg, velocity 33.3 m/s and elevation 3 m. The steam leaves the turbine at the following state: Pressure 20 kPa, enthalpy 2512 kJ/kg, velocity 100 m/s, and elevation 0 m. Heat is lost to the surroundings at the rate of 0.29 kJ/s. If the rate of steam flow through the turbine is 0.42 kg/s, what is the power output of the turbine in kW? (6) ②
- b) State Clausius inequality. (2)
4. a) State Charles' law. (1)
- b) What is PMM-I? Is it possible to devise a PMM-I? Justify your answer. $(1+1+1)$
- c) A house requires $2 \times 10^5 \text{ kJ/h}$ for heating in winter. Heat pump is used to absorb heat from cold air outside in winter and send heat to the house. Work required to operate the heat pump is $3 \times 10^4 \text{ kJ/h}$. Determine (i) Heat abstracted from outside and (ii) Co-efficient of performance. $(2+2)$
5. a) Derive the relationship between temperature and volume for an ideal gas, if it undergoes a polytropic process. <https://www.pyqonline.com> (2)
- b) A gas undergoes a thermodynamic cycle consisting of the following processes:
 - Process 1-2: Constant pressure $p = 1.4 \text{ bar}$, $V_1 = 0.028 \text{ m}^3$, $W_{12} = 10.5 \text{ kJ}$
 - Process 2-3: Compression with $pV = \text{constant}$, $U_3 = U_2$
 - Process 3-1: Constant volume, $U_1 - U_3 = -26.4 \text{ kJ}$. There are no significant changes in KE and PE.
 (i) Sketch the cycle on a p-V diagram (ii) Calculate the net work for the cycle in kJ. ①
 (iii) Calculate the heat transfer for process 1-2 (iv) Show that $\Sigma Q = \Sigma W$. $(1+2+1+2)$
6. a) Draw the phase equilibrium diagram on p-v & T-s coordinates considering water as a pure substance. Indicate all relevant constant property lines in the plot. (2+2)
- b) Find the specific enthalpy and specific entropy of steam when the pressure is 2 MPa and the specific volume is $0.09 \text{ m}^3/\text{kg}$. (2+2)
7. a) Derive the expression, $h = 353H \left[\frac{1}{T_a} - \left(\frac{m+1}{m} \right) \frac{1}{T_f} \right]$ of natural draught in m of water column. (all the symbols have their usual meanings). (5)
- b) Draw a neat sketch of a horizontal shell, water tube boiler and label its components. (3)
8. 2 kg of an ideal gas occupies a volume of 0.3 m^3 at 10 bar pressure and 500 K temperature. When this gas expands polytropically $pV^{1.2} = C$, the internal energy decreases by 300 kJ. Presuming adiabatic exponent $\gamma = 1.4$, determine: (i) characteristic gas constant, (ii) final temperature and pressure of gas and (iii) heat and work interactions across the system boundary. $[1+(2+2)+(2+1)]$
9. a) 10 kg of wet steam at a pressure of 0.2 MPa is contained in a rigid tank of volume 6.058 m^3 . The tank is heated until the steam becomes dry saturated. Determine the final pressure and the heat transfer to the tank. (3+3)
- b) What do you understand by degree of superheat? What is the value of pressure and temperature at

critical point in case of water?

(1+1)

10. a) Why accessories are essential in boiler? Name two accessories used in boiler and state the function of steam stop valve and fusible plug. ① (1+1+1+1)
- b) A boiler evaporates 5 kg of water per kg of coal burnt on the fire grate in the combustion chamber. The temperature of feed water is 30°C and the steam is raised at 10 bar absolute pressure. Calculate the equivalent evaporation "from and at 100°C " per kg of coal and the efficiency of boiler, if the steam produced is 0.95 dry. Take calorific value of coal as 29800 kJ/kg. (2+2)
-